



**UNIVERSITY OF
CALGARY**

ENGG 684- Introduction to Project Management

Project Charter

Group – 8 (M&M Constructions Ltd.)

Course Instructor:

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1. Statement of the Problem

M&M Constructions Ltd. has secured a significant business opportunity by being awarded a portfolio of eight diverse, multidisciplinary design projects. The complexity of this portfolio is evident in the unique demands of each project:

- Single-Family Residential Home – Functional space planning and cost-effective design.
- Low-Rise Office Building – Integrated mechanical and electrical systems.
- Mid-Rise Apartment Complex – Complex plumbing coordination and compatibility checks.
- Community Health Clinic – Safety, accessibility, and HVAC system design.
- Retail Strip Mall – Flexible layout and structural planning.
- Elementary School – Natural lighting, sustainability, and occupant safety.
- Underground Parkade – Geotechnical and structural challenges.
- Civic Library and Community Centre – Architectural aesthetics and robust building systems.

This diversity in the projects presents a substantial management challenge. Attempting to execute these projects in isolation, without a unified delivery strategy, exposes the company to the same pitfalls that lead to significant cost and schedule overruns on major projects. The primary problem is the risk of inadequate front-end loading (FEL) at the portfolio level. Without a robust, integrated plan, the company faces predictable and severe risks, including the creation of "overly optimistic original cost estimates and schedules," resource bottlenecks due to the high number of "interfaces and interdependencies," and cumulative scope creep. The core challenge is to develop a comprehensive project execution plan that proactively manages these portfolio-level complexities, ensuring the successful delivery of all eight projects within the highly constrained 40-week timeline.

2. Project Goal

To develop and implement a delivery plan with strategy that ensures the completion of all eight projects awarded to M&M Constructions Ltd. on time, within budget, and to the quality standards expected by stakeholders.

3. Scope of Planning (Objectives)

In scope	Out of scope
a) Strategic Framework Development b) Scheduling of all design tasks for the eight projects. c) Selection and management of subcontractors for each discipline. d) Cost Estimation and Financial Analysis. e) Risk identification and mitigation strategies. f) Performance Measurement (Defining the key performance indicators (KPIs), including Earned Value metrics (CPI, SPI), to monitor and control project performance during the simulation phase.)	a) The actual execution of design work (e.g. creating architectural drawings). b) Construction, physical procurement, or on-site management activities for the buildings, as our firm's mandate is strictly design engineering. c) Client-side project management activities, such as securing land permits or final building approvals. d) Direct contractual negotiations with the end-clients for each of the eight projects, we assume these contracts are already secured.

4. High-Level Schedule

The Overall Project Duration is: **40 weeks (maximum).**

Estimated Timeline of each project on weekly basis and the total duration of each project is following Projects

1: Week 1 – Week 7 (Total 7 weeks)

Projects 2: Week 3 – Week 13 (Total 10 weeks)

Projects 3: Week 8 – Week 19 (Total 11 weeks)
 Projects 4: Week 10 – Week 21 (Total 11 weeks)
 Projects 5: Week 12 – Week 24 (Total 12 weeks)
 Projects 6: Week 15 – Week 26 (Total 11 weeks)
 Projects 7: Week 19 – Week 30 (Total 11 weeks)
 Projects 8: Week 20 – Week 34 (Total 14 weeks)

Note: This schedule is an estimate and may vary during the actual progress of the work.

5. Resources

- **Internal Resources:** The Project Management team (group 8) is responsible for all strategic planning, analysis, decision-making, and reporting. Administrative staff are assumed to support logistical functions.
- **External Resources:** A list of pre-qualified subcontractors as provided, will serve as the primary execution workforce for the seven specialized design tasks. The selection and management of these external partners are a critical component of our strategy and will be performed by group 8 in further submissions.
- **Tools & Software:** A suite of project management tools will be utilized. MS Project will be used to build the critical path network and Gantt charts. Spreadsheets will be employed for cost estimation, Monte Carlo risk simulation, and cash flow analysis. The StratPM platform will be the core tool for simulating and evaluating the performance of the developed strategies.

6. Assumptions and Risks

Assumptions:

- The project data provided (work quantities, production rates, costs) is accurate for initial planning, forming the baseline for our estimates. **Implication:** Any significant deviation from this data in the simulation will test the robustness of our contingency planning.
- The pool of subcontractors listed is available, and their quoted production rates are achievable. **Implication:** A change in availability would represent a major external risk requiring immediate strategic adjustment.
- The 40-week deadline is a non-negotiable hard constraint. **Implication:** Strategies must be optimized for schedule adherence, even if it means incurring higher costs.
- All eight projects must be undertaken and completed; selective cancellation is not an option. **Implication:** The plan must be robust enough to accommodate all projects.

High-Level Risks:

- **Inadequate Front-End Loading:** Rushing the planning phase or creating an "overly optimistic original cost estimate and schedule" could lead to significant downstream cost and schedule erosion.
- **Resource Bottleneck:** Over-reliance on a single, high-performing subcontractor may create a critical path bottleneck, causing a cascading delay across the portfolio.
- **Cumulative Scope Creep:** The materialization of risks like "Unclear Client Requirements" or "Design Scope Creep" across several projects could have a cumulative impact that overwhelms the contingency plan.
- **Poor Subcontractor Performance:** Selecting subcontractors based solely on the lowest cost may introduce risks of lower productivity, rework, and delays, forcing costly adjustments.

7. Stakeholders

Stakeholder	Role / Interest & Analysis	Management Strategy
Design Company Owner	(Project Sponsor) A high-power, high-interest stakeholder who provides funding and defines ultimate business objectives.	Manage Closely: Provide regular, concise progress reports focusing on key financial and schedule KPIs. Ensure the final recommendation is directly aligned with

Stakeholder	Role / Interest & Analysis	Management Strategy
		their strategic goals for profitability and repeatability.
Project Management Team	(Group 8) A high-power, high-interest stakeholder responsible for all planning, analysis, and control.	Manage Closely: Maintain clear internal communication, defined roles, and a robust decision-making process to ensure alignment and effective execution of the project plan.
Subcontractors	External firms executing the design tasks. They are low-power, high-interest stakeholders in our planning, but their performance is critical.	Keep Informed: Develop clear work packages and communication protocols. While they have no say in our strategy, our plan's success depends entirely on their simulated performance.
End-Clients	(Implicit) The ultimate recipients of the designs. They are low-power, high-interest stakeholders in our planning context.	Keep Satisfied (Indirectly): While we have no direct contact, our strategy's success is ultimately measured by delivering designs that meet their needs on time and budget. Their satisfaction is a primary driver for the project's success criteria.

8. Criteria for Project Success

Success will be measured against the following specific and quantifiable criteria:

- **Schedule:** Completion of all eight project designs within the 40-week simulated timeframe.
- **Cost:** The final simulated total cost is within the planned budget, including contingency, and the project concludes with a positive cash balance.
- **Quality & Scope:** All seven design tasks for all eight projects are completed with minimum penalty related to rework or quality issues during the simulation.
- **Strategic Recommendation:** The final recommendation in the Close-Out Report is data-driven, well-justified, and demonstrates a clear, quantitative analysis of the trade-offs between different management strategies.

AI Critical Assessment and Final Version Justification:

Our final version of the Project Charter is very different from initial AI responses because the AI is trained on the data which we provided which included Course Project_V2 and V3 document but that has only information related to the project and AI is giving a vague answer or the response which is correct on the surface level and the response which we needed was according to what professor detailed us during the lecture and even we also tried to put the lecture detail to the AI but it doesn't know what professor communicated to us and for that we have also tried to tell AI what we want in detail and then AI is giving us the response which is very close to what we wanted but still there were some things which are out of the contexts and then we manually made the project charter on AI's response and what we required, we have to delete a lot of stuff which AI provided because it was unnecessary for us, at least in the project some examples are: when we asked AI about statement of the problem AI response was good but then it was focusing on duty and challenges and our group's focus was to get something on front end loading which professor explained during class but we weren't receiving answers in that direction and then we have to modify the statement and similarly we have faced problem with AI in Schedule the AI response was on planning in different phases but what we wanted is how we are dividing each project week wise then we manually plotted everything in excel and taking consideration of Contractor C in each task and then calculated a high level schedule but AI's response was not correct or at least not what we want in our charter so this is how we have taken a lot of ideas from AI and then implemented using our course work requirements accordingly.

APPENDIX I: AI Log

Following is the AI log for Project charter, we used ChatGPT and DeepSeek for the same:

Question: can you please define the project charter

AI response: The project charter is a foundational, formal document that officially authorizes a project or a phase of a project. It's essentially the birth certificate of the project and serves as a high-level agreement between the project manager, the sponsor, and the key stakeholders. Here's a breakdown of its key purposes and common components:

Key Purposes of a Project Charter

Authorization: It officially sanctions the existence of the project and gives the project manager the authority to apply organizational resources to project activities.

Stakeholder Buy-in: It confirms the commitment of the organization to the project and aligns stakeholders on the project's goals, scope, and objectives.

High-Level Vision: It provides a common understanding of the project's objectives and boundaries at a high level before detailed planning begins.

Initial Scope Definition: It establishes the preliminary boundaries of the project and identifies what is in and out of scope.

Common Components of a Project Charter

While the exact content can vary by organization, a typical project charter includes:

Project Title and Date: Simple identification.

Project Purpose or Justification: A brief statement explaining the problem the project will solve or the opportunity it will address (the "Why").

High-Level Project Description: A summary of the project and what it entails.

Measurable Project Objectives and Success Criteria: What constitutes a successful outcome (e.g., "Reduce customer complaints by 15% within six months of launch").

High-Level Requirements: Broad statements describing what the product, service, or result must achieve.

Overall Project Risks: Initial identification of major threats or opportunities.

Summary Milestone Schedule: A very high-level timeline showing key deliverables and their target dates.

Summary Budget: The authorized financial resources for the project.

Key Stakeholder List: Identification of the main people or groups affected by the project.

Project Approval Requirements: Who signs off on the final project outcome.

Assigned Project Manager: The name and authority level of the person leading the project.

Name and Signature of the Sponsor: The signature of the person or body authorizing the project and funding it.

In short, the project charter is the foundational document that sets the stage for everything else. No project should move into detailed planning or execution until the charter has been formally approved.

Question: Hi gpt, So we have a Construction company known as M&M Constructions Ltd. has been given projects including residential commercial, educational, healthcare, and community buildings, each requiring architectural, geotechnical, structural, electrical, mechanical, plumbing, and compatibility tasks. We want a problem of statement for that, can you please explain that in detail.

AI response:

Comprehensive Project Overview

1. Project Portfolio & Strategic Context

M&M Constructions Ltd. is tasked with the simultaneous delivery of eight distinct, multidisciplinary design projects. This portfolio represents a significant strategic undertaking, characterized by diverse requirements and specialized challenges:

Single-Family Residential Home: Prioritizing functional space planning and cost-effective design.

Low-Rise Office Building: Centered on the integration of complex mechanical and electrical systems.

Mid-Rise Apartment Complex: Demanding advanced plumbing coordination and multi-trade compatibility checks.

Community Health Clinic: Governed by stringent requirements for safety, accessibility, and specialized HVAC design.

Retail Strip Mall: Requiring a flexible structural and layout plan to accommodate future tenant variability.

Elementary School: Focusing on natural lighting optimization, sustainable materials, and occupant safety.

Underground Parkade: Involving significant geotechnical and structural engineering challenges.

Civic Library and Community Centre: Balancing prominent architectural aesthetics with the deployment of robust, high-usage building systems.

Each project necessitates a full suite of design disciplines: Architectural, Geotechnical, Structural, Electrical, Mechanical, Plumbing, and final cross-disciplinary Compatibility Checks.

2. The Strategic Imperative and Inherent Risks

The primary challenge is the coordinated execution of this entire portfolio within a constrained 45-week timeline. The absence of a unified, master-program-level strategy will expose the company to a cascade of inefficiencies, including:

Schedule Overlaps and Conflicts: Critical resources, such as structural engineers or BIM coordinators, being scheduled for multiple projects simultaneously.

Resource Bottlenecks: Scarcity of specialized personnel or subcontractors leading to work stoppages and cascading delays.

Subcontractor Misalignment: A lack of synchronized information flows and schedules, resulting in site conflicts, rework, and contractual disputes.

Critical Path Ambiguity: Failure to identify and manage the inter-project dependencies, making the entire 45-week program vulnerable to delays from any single project's setback.

Budget Erosion: The direct financial consequence of the above: idle time, premium-rate rush orders, and corrective work leading to significant cost overruns.

3. Objective: Defining the Delivery Framework

The planning stage must therefore establish a clear, integrated, and resilient project framework. This framework is designed to synchronize all moving parts, optimize resource allocation across the portfolio, and enforce rigorous timeline and cost controls. The ultimate goal is to guarantee the simultaneous, on-time, on-budget, and high-quality delivery of all eight projects.

4. Scope & Focus: Related Unwanted Topics for this Planning Stage

To maintain focus and efficiency during the strategic planning phase, the following topics, while related to construction and project management, are considered out of scope for this specific framework

establishment and will be addressed in subsequent, more detailed phases. The current discussion will explicitly avoid:

Detailed Architectural Schematics: While overall space planning is a requirement, deep dives into specific aesthetic choices, material palettes, or facade details are for later, project-specific design charrettes.

Subcontractor Prequalification and Bidding: The assumption is that qualified subcontractors are available. This planning focuses on how to manage them, not which ones to select.

Micro-level Task Scheduling for Individual Trades: The framework will define major milestones and interdependencies (macro-scheduling), not daily or weekly work plans for electricians or plumbers.

Definitive Budgetary Calculations: The focus is on the cost-control mechanism and risk-mitigation strategies, not on establishing the final, detailed budget for each project line item.

Resolution of Specific Geotechnical or Structural Engineering Problems: For example, the exact piling solution for the Parkade. The framework ensures the process for solving these challenges is robust, but the technical solution itself is for the engineering teams to develop.

Marketing, Public Relations, and Community Engagement Plans: These are critical for project success but are separate workstreams from this core operational and delivery framework.

Legal and Contractual Fine-Print: The framework assumes standard contracts will be used; legal nuances are for the company's counsel to review separately.

Long-term Facility Maintenance and Lifecycle Costing: While sustainability (e.g., in the School) is a design goal, the operational planning for post-construction maintenance is beyond the 45-week delivery scope.

Question: What shall be statement of the problem for project charter?

AI Response: Of course. Based on the project description, here is a strong, comprehensive Statement of the Problem for your Project Charter.

This statement answers the two key questions posed in the brief:

1. Why does the company need a clear delivery strategy?
2. What challenges must be addressed in the planning stage?

Statement of the Problem

Our design engineering company is tasked with the simultaneous delivery of eight multidisciplinary design projects within a strict 45-week timeframe. The lack of a unified, strategic delivery plan poses a significant threat to the company's success, as the current ad-hoc approach is insufficient to manage the project portfolio's complexity. Without a clear strategy, we face several critical challenges that must be proactively addressed in the planning stage:

1. **Resource Constraints and Coordination Complexity:** Managing seven specialized design tasks (Architectural, Geotechnical, Structural, Electrical, Mechanical, Plumbing, and Compatibility Check) across eight distinct projects requires meticulous coordination of limited subcontractor resources. Without an optimized schedule, we risk severe resource conflicts, idle time, and an inability to meet the aggressive deadline.
2. **Financial Viability and Cash Flow Risk:** The projects have fixed revenues, but variable costs based on subcontractor selection and project duration. The company is exposed to potential negative cash flow due to the timing of significant upfront subcontractor payments versus receiving client payments only upon project completion. Without careful financial planning and risk analysis, the entire portfolio could become unprofitable.
3. **Scope and Performance Variability:** Each of the eight projects has a unique scope, complexity, and client requirements (e.g., the Civic Library's focus on aesthetics vs. the Underground Parkade's geotechnical challenges). A one-size-fits-all approach will lead to performance gaps, scope creep, and failure to meet specific client expectations.
4. **Unmanaged Risk Exposure:** The design phase is susceptible to numerous cost-related risks, including design errors, regulatory changes, subcontractor delays, and scope creep. Without a formal risk management process to identify, analyse, and mitigate these threats, the project's budget and schedule are highly vulnerable.

Therefore, a comprehensive delivery strategy is not just beneficial but essential to navigate these interconnected challenges of time, cost, resources, and risk, ensuring the successful and profitable delivery of all eight projects.

Question: The statement of the problem shall also include the name of the 8 projects, not the vague one, please give a detailed response?

AI response: Statement of the Problem

Project: "Managing a Design Company" - ENGG 684 Course Project

Our Role: We are the management team of a design engineering company responsible for the end-to-end delivery of the design phase for a portfolio of eight distinct building projects.

The Core Problem: Our company lacks a defined, integrated strategy to manage the simultaneous delivery of eight multidisciplinary design projects within the strict 45-week deadline. Our current ad-hoc approach is inadequate to handle the complexity and interdependencies of the tasks, putting us at high risk of missing deadlines, exceeding budgets, and failing to meet client expectations.

Our Specific Duties & The Associated Challenges:

As the management team, our primary duty is to create and execute a delivery strategy. This involves making critical decisions in the following areas, each presenting a significant challenge:

Subcontractor Management & Scheduling:

Duty: Select and coordinate specialized subcontractors for each of the seven tasks (Architectural, Geotechnical, Structural, Electrical, Mechanical, Plumbing, Compatibility Check) across all eight projects.

Challenge: Optimizing their selection and sequence to avoid resource conflicts and complete all work within 45 weeks, given their varying production rates and costs.

Financial Planning & Cash Flow Control:

Duty: Develop accurate cost estimates, establish a contingency fund based on project risks, and manage weekly cash flow.

Challenge: Preventing negative cash flow due to upfront subcontractor payments while client payments are received only upon project completion, which could lead to insolvency.

Risk Mitigation:

Duty: Proactively identify, analyze, and mitigate cost-related risks (e.g., redesign, regulatory changes, delays) that could impact the project's budget and schedule.

Challenge: Quantifying the financial impact of these risks and deciding whether to accept them or pay for insurance, without eroding the project's profitability.

Scope & Quality Assurance:

Duty: Ensure that each of the eight unique projects (from a Single-Family Home to a Civic Library) meets its specific design requirements and quality standards.

Challenge: Managing the diverse scopes and complexities without allowing scope creep or quality issues to cause delays and cost overruns.

Therefore, the fundamental problem we must solve in this planning stage is to integrate these duties into a coherent, resilient, and financially-sound master plan that will guide our company to the successful and profitable completion of all eight projects.

Question: the statement shall be crisp and shall include "We must deliver eight diverse design projects within 40 weeks, each requiring coordinated work across architecture, structural, mechanical, electrical, and plumbing disciplines. The risk is that without a unified strategy, we will face scheduling conflicts, misaligned teams, and cost overruns, jeopardizing timely delivery. The challenge is to create a single, efficient project framework that ensures all eight projects are delivered on time, on budget, and to quality standards."

AI Response: Statement of the Problem

We must deliver eight diverse design projects within 45 weeks, each requiring coordinated work across seven specialized disciplines: Architectural, Geotechnical, Structural, Electrical, Mechanical, Plumbing, and Compatibility.

The risk is that without a unified strategy, we will face severe scheduling conflicts, resource inefficiencies, and cost overruns, jeopardizing timely delivery and profitability.

The challenge is to create a single, efficient project framework that aligns our resources and processes to ensure all eight projects are delivered on time, on budget, and to quality standards.

Question: What shall be the project goal?

AI Response: Project Goal

To develop and implement an integrated project delivery strategy that successfully completes the design for all eight multidisciplinary projects within the 45-week deadline, while optimizing resource utilization, mitigating financial risk, and ensuring overall project profitability.

Breaking Down the Goal:

This goal is effective because it is:

Strategic and High-Level: It doesn't get bogged down in specific tactics but sets the ultimate direction for your company.

Comprehensive: It addresses the three critical constraints of any project, known as the "Iron Triangle":

Time: "...within the 45-week deadline."

Cost: "...ensuring overall project profitability."

Scope: "...successfully completes the design for all eight multidisciplinary projects..." (implied quality and scope).

Action-Oriented: It uses powerful verbs like "develop," "implement," and "optimize," which reflect proactive management.

Aligned with the Problem: It directly responds to the challenges identified in the Problem Statement—scheduling complexity, financial risk, and resource coordination.

Question: What shall be the scope of planning (Objectives) for the project charter and please make sure it should include “framework to coordinate the multidisciplinary design of eight projects within 45 weeks, focusing on integrated scheduling, resource allocation, financial control, risk mitigation, and performance monitoring, without involving detailed engineering design or physical construction.”

AI Response: Scope of Planning (Objectives)

This plan defines the strategic framework for delivering the eight design projects. Our focus is on project management and delivery strategy, not on producing the actual technical designs.

Included in Planning (Our Objectives):

Schedule Development: To develop a detailed, integrated master schedule (Gantt chart) that sequences all tasks and subcontractors, ensuring the completion of all eight projects within the 45-week constraint.

Resource & Subcontractor Strategy: To select and allocate specialized subcontractors for each of the seven disciplines per project, optimizing for a balance of cost, speed, and reliability.

Financial Framework: To create a comprehensive cost estimate, establish a contingency fund based on quantitative risk analysis (Monte Carlo simulation), and model cash flow to ensure financial viability.

Risk Management Plan: To identify, analyze, and develop mitigation strategies (including insurance purchase decisions) for the cost-related risks provided in the project brief.

Performance Monitoring & Control: To define a process for tracking project performance using Earned Value Analysis (EVA) at key milestones (Weeks 10, 20, 30, 45) and making necessary adjustments during simulation.

Final Recommendation: To simulate the execution of our strategies, compare outcomes, and synthesize our findings into a professional recommendation for the most effective overall delivery strategy.

Not Included in Planning (Exclusions):

Detailed Engineering Design: We are managing the design process, but not creating the architectural drawings, structural calculations, or other technical deliverables ourselves.

Physical Construction: This project is solely about the design phase. Construction, site supervision, and commissioning are out of scope.

HR & Staffing Functions: We are selecting subcontractors as entities. Hiring, firing, or managing the internal payroll of individual employees is excluded.

Marketing & Business Development: Acquiring new projects or client negotiations beyond the given eight are not part of this plan.

Definitive Legal Contracts: While we consider subcontractors, we will not be drafting detailed legal agreements or negotiating contract clauses.

Software Development: The management and use of software tools (like MS Project, StratPM) are in scope, but the development of new software platforms is excluded.

Question: can you please define the High-level schedule for project charter?

Answer:

High-Level Schedule

The overall project must be completed within a 45-week deadline. The following timeline outlines the key phases and major milestones for the delivery of the eight design projects.

Phase	Major Milestones / Key Activities	Estimated Timeframe
Phase 1: Strategic Planning & Setup	Finalize Project Charter; Initial Subcontractor Selection; High-Level Strategy Formulation.	Weeks 1 - 3
Phase 2: Detailed Scheduling & Mobilization	Develop Detailed Master Schedule (Gantt Chart); Finalize Subcontractor Contracts; Financial Modeling & Cash Flow Setup.	Weeks 4 - 6
Phase 3: Concurrent Project Execution	Execute all eight design projects in parallel streams. • Key Checkpoints: Project Reviews at Week 10, 20, and 30. • Ongoing: Subcontractor coordination, quality checks, and financial monitoring.	Weeks 7 - 40
Phase 4: Project Closeout & Integration	Final Compatibility Checks; Project Deliverable Submission; Client Handover; Final Payments.	Weeks 41 - 43
Phase 5: Review & Reporting	Internal Performance Review; Lessons Learned Analysis; Final Report and Recommendation.	Weeks 44 - 45

Overall Project Duration: 45 Weeks

Question: That's a good schedule but we want it to be divided in a way that for project 1 it will take 11 weeks and for project 2 it will take 9 weeks and the list goes on, because in the lecture professor discussed this is how we are supposed to give the high level schedule, not as you are describing.

AI Response:

High-Level Schedule

The following table outlines the planned duration for the design and compatibility check of each of the eight projects. These projects will be executed in a concurrent, overlapping manner to meet the overall 45-week deadline.

Project Name	Planned Duration (Weeks)	Key Milestones (e.g., Major Discipline Completion)
Project 1 – Residential Home	11 weeks	Architectural & Structural Sign-off (Week 6)
Project 2 – Office Building	9 weeks	Mechanical & Electrical Coordination (Week 5)
Project 3 – Apartment Complex	12 weeks	Plumbing & Compatibility Check (Week 10)
Project 4 – Health Clinic	10 weeks	HVAC & Safety Review (Week 7)
Project 5 – Retail Strip Mall	9 weeks	Structural & Layout Finalization (Week 5)
Project 6 – Elementary School	11 weeks	Sustainability & Safety Review (Week 8)
Project 7 – Underground Parkade	10 weeks	Geotechnical & Structural Sign-off (Week 6)
Project 8 – Library & Community Centre	13 weeks	Architectural Aesthetics & Systems Integration (Week 9)
Total Programme Duration	45 weeks	Final Portfolio Delivery (Week 45)

Question: What shall be the resources for our project which we can add in project charter?

AI Response:

To successfully deliver the eight design projects, our company will utilize a combination of internal team members, software tools, financial assets, and specialized external subcontractors.

1. Internal Project Team

Project Management Team (Our Group): Responsible for strategic planning, scheduling, cost estimation, risk management, and overall project control.

Administrative Staff: Handles procurement, contracts, payments, and general office support.

2. Software & Tools

Scheduling Software: Microsoft Project for developing and maintaining the detailed master schedule and Gantt charts.

Simulation Platform: StratPM for simulating project execution and testing our strategy.

Financial Modeling Tools: Microsoft Excel for cost estimating, Monte Carlo risk simulation, and cash flow analysis.

Collaboration Tools: D2L and shared cloud storage for team coordination and document version control.

3. Financial Resources

Initial Mobilization Payment: \$500,000 to fund startup costs.

Project Revenue: Total potential revenue of \$3,200,000 from the eight projects, disbursed upon each project's completion.

Contingency Fund: A separately managed budget derived from quantitative risk analysis to address unforeseen cost events.

4. External Subcontractors

We will engage specialized external firms for each of the seven design disciplines. Our strategy will involve selecting from the following pool to optimize for cost, speed, and reliability:

Architectural Design

Geotechnical Engineering

Structural Engineering

Electrical Engineering

Mechanical Engineering (HVAC)

Plumbing Design

Compatibility Check

The final selection for each task and project will be determined during the detailed scheduling phase (Segment 2), based on a strategy that balances project duration and total cost.

Question: Who all will be our stakeholders for the project?

Stakeholders

Internal Stakeholders

Project Management Team (Our Group): Directly responsible for the planning, execution, and success of the project delivery strategy.

Company Owner / Senior Management: Provides overall strategic direction, approves the final project strategy and budget, and is the ultimate decision-maker. (This is the recipient of your Professional Recommendation Letter in Segment 4).

Administrative & Finance Staff: Supports contract management, payment processing to subcontractors, and tracking of indirect costs.

External Stakeholders

Clients (for each of the 8 projects): The entities who have commissioned the designs. They define the requirements and will accept the final deliverables. Their satisfaction is a primary measure of success.

Subcontractors: The specialized firms who perform the actual design work. Their performance, reliability, and cost are critical to our schedule and budget.

Regulatory Authorities (e.g., City Planning, Building Code Officials): Define the codes and standards that the designs must comply with for approval.

Insurance Providers: Offer policies to mitigate specific financial risks identified in the risk register (e.g., design errors, third-party delays).

Question: And what can be our criteria for our success?

AI Response: Criteria for Project Success

The success of our project delivery strategy will be measured against the following key performance indicators (KPIs):

On-Time Delivery:

Success: All eight design projects are 100% completed within the 45-week total project duration.

Measurement: Master schedule and StratPM simulation results show zero projects exceeding their planned finish date.

Financial Performance:

Success: The total project cost (Direct + Indirect + Contingency) remains below the total revenue of \$3,200,000, resulting in a net profit.

Measurement: Final cost analysis shows a positive cash flow and a profit margin upon completion of all projects.

Budget Adherence:

Success: The final cost does not exceed the initial budget plus the allocated contingency fund.

Measurement: Actual costs tracked against the estimated budget, with contingency usage documented and justified.

Resource Efficiency:

Success: Subcontractors are utilized effectively with minimal idle time, and no critical resource conflicts cause project delays.

Measurement: Gantt chart and simulation outputs show efficient resource leveling and on-time task completion.

Quality & Scope Fulfillment:

Success: All seven design tasks for each of the eight projects are completed and pass the final compatibility check, meeting the defined scope and quality standards.

Measurement: Zero projects are rejected due to compatibility failures or missing design disciplines.

Strategic Decision-Making:

Success: The strategy proves resilient, allowing for effective adjustments during execution based on Earned Value Analysis (EVA) at checkpoints.

Measurement: Successful mid-course corrections in the StratPM simulation that keep projects on track.